

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

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Homework 20

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Explain the key differences between shared memory systems and distributed memory systems, including how memory is accessed and how processes communicate, and discuss one advantage and one disadvantage of each model.
2. Describe the characteristics of processes in MPI programs, including how many processes exist, how they execute programs, and how they are identified, and explain why MPI programs follow the SPMD model.
3. Explain the purpose of `MPI_Init` and `MPI_Finalize`, including why no MPI calls can occur before or after these functions, and describe what could go wrong if this rule is violated.
4. Explain the role of communicators in MPI, define `MPI_COMM_WORLD`, and describe how `MPI_Comm_size` and `MPI_Comm_rank` are used in a parallel program.
5. Explain how `MPI_Send` and `MPI_Recv` work together, including the meaning of parameters such as destination, source, tag, and communicator, and describe how message matching occurs.
6. Explain how a process can receive a message without knowing the sender, message size, or tag in advance, and describe the role of `MPI_Status` and `MPI_Get_count` in this process.
7. Discuss the potential issues with `MPI_Send` and `MPI_Recv`, including blocking behavior and implementation-dependent behavior, and explain how these issues can affect program correctness and performance.

- **Welcome: Sean**

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